Year 9 Science – Booklet 2: Chemistry

Bonding & Structure — ANSWERS

Quick Test (p.3)

1. An element in its atomic form – separate, unjoined atoms of one element (not paired into molecules).
2. Molecules of an element – atoms of the same element joined together in pairs.
3. Molecules of a compound – atoms of two different elements chemically joined together.
4. A gas.
5. A thermometer.
6. Reactants.
7. Products.
8. The product (iron sulphide) is a black, non-magnetic solid – very different from the magnetic iron metal and yellow sulphur it was made from, and no longer attracted to a magnet, showing a new substance with different properties has formed.
9. Yes – a compound always has a fixed composition (a set ratio of its elements), e.g. iron sulphide (FeS) always has one iron atom for every one sulphur atom.
10. No – the elements in a compound are chemically joined together and cannot be separated by simple physical means (unlike a mixture), e.g. iron sulphide cannot be separated back into iron and sulphur using a magnet.

(Q1–3 refer to the small particle-diagram icons in the Quick Test box – check the colouring/grouping in your copy matches this reading: single scattered dots = element, paired same-colour circles = molecules of an element, paired different-colour circles = molecules of a compound.)

Section A – Multiple Choice (5 marks)

1. c) it's a yellow solid
2. b) it's a magnetic metal
3. d) it's a compound
4. a) use a magnet
5. a) it cannot be easily separated

Section B (15 marks)

1. Anagrams & definitions (5 marks)

• **cduopomn** → **compound** — a substance which is made of atoms of two or more elements that have been chemically joined together

• **yeproprt** → **property** — a quality which is always present in a substance, however much of it is present

• **eeculomc** → **molecule** — a small piece of a substance which has all the properties of the substance

• **mentag** → **magnet** — a substance which repels another magnet

• **icatreno** → **reaction** — a chemical change

2. Complete the passage (10 marks)

Elements are made of only one type of **atom**. If atoms of the same elements are joined together they form **molecules** of the element. The oxygen and nitrogen in the air around us are in the form of molecules.

If the atoms of two (or more) different elements are chemically joined together they form molecules of a **compound**. New substances are formed by chemical **reactions**. These new substances are called the **products**. The products of a chemical reaction can have very different properties from the **reactants** at the start of the reaction. When iron and sulphur are heated together they can form a new substance called **iron sulphide**. Iron is a magnetic **metal**. Sulphur is a **yellow** solid. But, **iron sulphide** has very different properties from both iron and sulphur – it is a non-magnetic black solid.

Blanks in order: atom · molecules · compound · reactions · products · reactants · iron sulphide · metal · yellow · iron sulphide.

Section C (4 marks)

1. How could Nikesh tell that a chemical reaction was taking place? (1 mark)

Bubbles of gas would be seen forming in the flask (effervescence), and/or the reading on the mass balance would decrease as the carbon dioxide gas produced escapes into the air.

2a) i) What else can have caused the decrease in mass? (part of 2 marks)

The carbon dioxide gas produced by the reaction escaping from the flask into the air (gas has mass, so the flask + contents weigh less once it has left).

2a) ii) Name one of the reactants in Nikesh's experiment.

Magnesium carbonate (or hydrochloric acid).

2b) Which diagram (A, B, C or D) best represents the carbon dioxide gas produced? (1 mark)

C

A gas is represented by particles that are far apart and randomly arranged, moving freely in all directions – unlike a solid (particles packed tightly in a regular arrangement, as in A and D) or particles bunched close together. As carbon dioxide is a compound, its particles are shown as joined molecules (one carbon atom bonded to two oxygen atoms) spread far apart, which is diagram C. If your copy of the booklet shows the diagrams differently, check that your answer is the option showing molecules spread far apart with empty space between them.